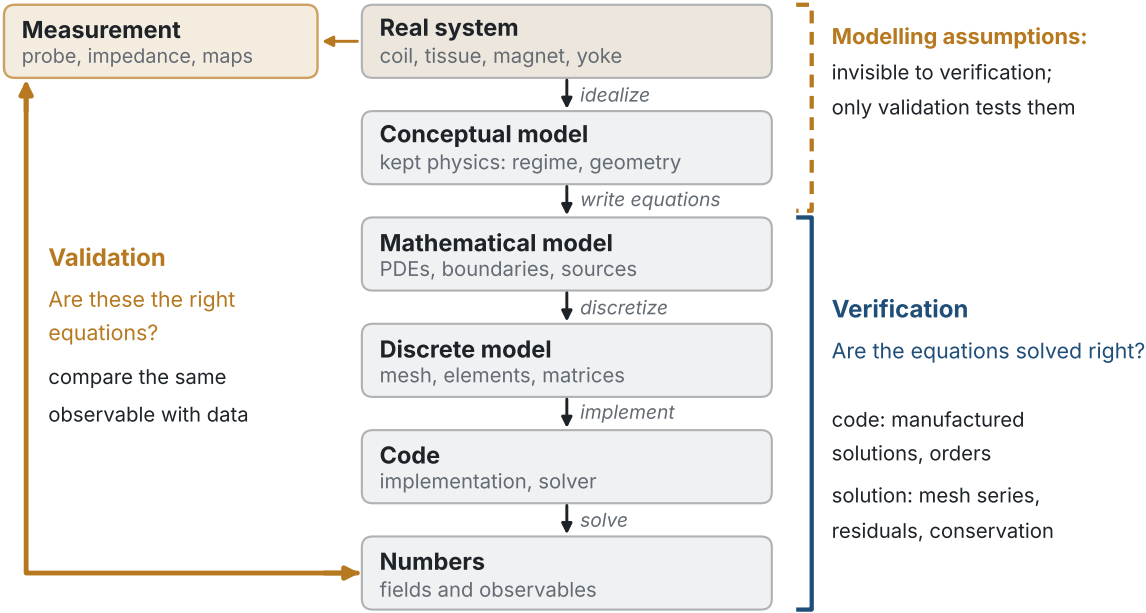
Measurement
probe, impedance, maps
Real system
coil, tissue, magnet, yoke
idealize
Conceptual model
kept physics: regime, geometry
write equations
Mathematical model
PDEs, boundaries, sources
discretize
Discrete model
mesh, elements, matrices
implement
Code
implementation, solver
solve
Numbers
fields and observables
Modelling assumptions:
invisible to verification;
only validation tests them
Validation
Are these the right equations?
compare the same observable with data
Verification
Are the equations solved right?
code: manufactured solutions, orders
solution: mesh series, residuals, conservation